



NIELIT Virtual Academy

National Institute of Electronics and Information Technology, Chennai

Autonomous Scientific Society of Ministry of Electronics & Information Technology (MeitY), Govt. of India,

ISTE Complex, 25, Gandhi Mandapam Road, Chennai – 600025

Course Prospectus

NSQF Aligned

Mode: ONLINE (Blended)

Artificial Intelligence Development Associate

Jointly developed by NIELIT and INTEL INDIA



Index

Topic	Page No.
Objective of the Course.....	3
Outcome of the Course.....	4
Course Structure	4
Course Fee Structure	5
Registration Fee	5
Eligibility.....	5
Number of Seats.....	6
How to Apply	6
Registration.....	6
Selection Criteria of candidates.....	6
Admission.....	7
Discontinuing the course.....	7
Important Dates.....	8
Examination & Certification.....	8
Grading Scheme.....	9
Detailed curriculum.....	9

Course Prospectus

Course Name : Artificial Intelligence Development Associate

Course Code: DS191

NSQF Level: 4

Duration: 570 Hours

Last Date of Registration: 04-06-2025

Date of publishing Provisional Selection List: 06-06-2025

Last Date of Payment of 1st Installment fee: 08-06-2025

Course Start Date: 09-06-2025

Preamble:

In the era of intelligent systems and data-driven innovation, the role of the **AI Development Associate** is more vital than ever. As industries increasingly adopt artificial intelligence to optimize operations, enhance decision-making, and create smarter user experiences, the need for professionals equipped with foundational AI development skills continues to surge. This program is designed to prepare aspiring developers with the essential knowledge and practical capabilities required to contribute effectively to AI projects.

Blending core principles with hands-on application, the AI Development Associate pathway focuses on areas such as **machine learning, deep learning, natural language processing, and computer vision**. Participants will also gain exposure to modern tools and frameworks used to build and deploy AI-powered solutions. Whether supporting data science teams, building intelligent features, or automating decision workflows, the AI Development Associate serves as a crucial link in the growing AI ecosystem.

Objective of the Course:

The objective of the AI Development Associate program is to **equip learners with foundational knowledge and practical skills** required to design, build, and deploy intelligent applications using artificial intelligence technologies. This program aims to:

- Introduce key concepts in AI, including machine learning, deep learning, NLP, and computer vision.
- Develop hands-on proficiency with AI tools, libraries, and frameworks.
- Enable learners to solve real-world problems using data-driven approaches.

- Prepare participants for entry-level roles in AI development and further specialization in the AI and data science domain.

Outcome of the Course:

After completion of the module, the students shall be able to:

- Create deployable AI models addressing real-world problems effectively and efficiently.
- Gain proficiency in Python, Orange, Tableau, and advanced AI libraries for industry applications.
- Process, evaluate, and extract meaningful insights from structured and unstructured data sources.
- Design fair, transparent, and unbiased AI systems considering societal and ethical implications.
- Cultivate an entrepreneurial mindset to innovate AI-driven solutions for emerging industrial challenges

Course Structure:

Module No.	Module Name	Theory	Practical	Total
1.	Implementation of Basic AI Solution using Python programming language and SMART Framework.	15	15	30
2.	Solving use cases using AI models along with building up Entrepreneurial Mindset	25	35	60
3.	Realization of Projects in AI domains with an understanding of AI Project Pitfalls	70	80	150
4.	Solving Real-time industrial problem statements using AI	10	20	30
5.	Employability Skills	-	-	60
6.	Implementation of AI project in virtual environment/ OJT	-	-	240
	Duration (in Hours)	120	150	570

Course Fee Structure:

	General/OBC/EWS	SC/ST	Last Date
Registration Fee	Rs.1000/-	NIL	04-06-2025
Tuition Fee (Including NSQF Registration & Examination Fee)			
1 st Installment	Rs.12,500/-	NIL	08-06-2025
2 nd Installment	Rs.12,500/-	NIL	09-08-2025
Total	Rs 26000/-	NIL	

Registration Fee- Refund Policy:

(Non-Refundable if candidate is selected for admission but did not join and if a candidate has applied but not eligible.)

However, the registration fee shall be refunded on few special cases as given below:

- 1) Candidates are eligible but not selected for admission.
- 2) Course postponed and new date is not convenient for the student.
- 3) Course cancelled.

Eligibility :

<i>Minimum Educational Qualification and Experience</i>	Pursuing the 1st year of UG in any discipline OR 12th Grade Pass OR Completed 2nd year of the 3-year diploma in CS/IT/EC/EE/ allied after 10 th OR 12th Grade Pass who have undergone the ARTIFICIAL INTELLIGENCE course as a Vocational subject in class 12 th ^ OR Previous relevant Qualification of NSQF Level 3.0 with a minimum education of 10th Grade pass with 3 years of relevant experience.
	^ Such students shall be exempted from the first 2 NOSs.

Number of Seats: 30 – Total

Note: Seats are allocated based on the merit of the Qualification

How to Apply?

Candidates can apply online in our **website <http://nva.nielit.gov.in>**. Payment towards non-refundable registration fee can be paid through any of the following modes:

- ✓ **Payment Gateway**
- ✓ **Online transaction: Account No: 31185720641 Branch: Kottur (Chennai), IFSCCode: SBIN0001669.**
- ✓ **GPAY/any UPI, Credit Card**

Note: The Institute will not be responsible for any mistakes done by either the bank concerned or by the depositor while remitting the amount into our account

Last date of Registration: 04-06-2025

Registration Procedure

All interested candidates are required to fill the Registration form online with registration fees of Rs. 1,000/- (wherever applicable) and with all the necessary information.

Selection Criteria :

Selection of candidates will be based on their marks in the qualifying examination subject to eligibility and availability of seats.

- ✓ The first list of Provisionally Selected Candidates will be published on NIELIT Chennai website (www.nielit.gov.in/chennai/index.php) on **06-06-2025 by 5:00 PM.** In case of vacancy, an additional selection list will be prepared and the selection will be intimated by email only.
- ✓ Following documents of candidates will be verified:
 - Qualifying Degree (Consolidated Marksheet/Degree Certificate/Course Completion Certificate), 10th and 12th mark sheet.
 - One passport size photograph.
 - Self-attested copy of Govt. issued photo ID card.
 - AADHAR Copy
- ✓ All provisionally selected candidates have to pay first instalment of Rs. **12,500/-** on or before **08-06-2025** by payment mode mentioned above.
- ✓ Selected candidates are requested to upload the proof of remittance of fee on registration portal and also send the proof of remittance of fee as email to **trng.chennai@nielit.gov.in / sourav.a@nielit.gov.in**

Admission:

All provisionally selected who have paid the fees (full or first instalment) and verified by accounts section of NIELIT Chennai will get a welcome message in his login id provided during registration.

Note-All Provisionally Selected Candidates have to visit NIELIT Chennai for Certificate Verification.

Otherwise their candidature will be cancelled without any intimation.

The credentials and URL for online portal will be shared through WhatsApp or email.

Discontinuing the course

- ✓ No fees under any circumstances, shall be refunded in the event of a student who have completed the process of admission or discontinuing the course in between. No certificate shall be issued for the classes attended.
- ✓ If candidates are not uploading consecutive 3 assignments within assigned time, then their candidature will be cancelled without any notice and all fees paid will be forfeited.
- ✓ If candidates are not appearing for any internal examinations/practical their candidature will be cancelled without any notice and all fees paid will be forfeited.

Course Timings:

This program is a practical oriented one and hence there shall be more lab than theory classes. The classes and labs are online cloud-based from **10 am to 1:30 pm** and Monday to Friday.

Course enquiries

Students can enquire about the various courses either on telephone or by personal contact between 9.15 A.M. to 5.15 P.M. (Lunch time 1.00 pm to 1.30 pm) Monday to Friday.

Placement:

Students who have completed the course successfully and qualified, Placement guidance and career counselling will be given to assist in their interviews.

Important Dates

Last Date of Registration: 04-06-2025

Display of Provisional Selection List: 06-06-2025

Payment of first installment fee: 08-06-2025

Course Start Date: 09-06-2025

Payment of second instalment fee: 09-08-2025

Examination & Certification

- ✓ Final Certificates will be issued after successful completion of all the modules including mini project. For getting certificate a candidate has to pass each module individually with minimum required marks of 50%.

NSQF Examination Pattern:

➤ Means of assessment:

S. No	Examination Pattern	Modules Covered	Duration in Minutes	Maximum Marks
1.	Theory Paper-1:	Module 1,2,4	90	100
2	Theory Paper-2:	Module 3	90	100
4	Practical 1:	Module 1,2,3,4	180	90
6	Internal Assessment	Employability Skill	-	30
8	Project	Implementation of AI project in virtual environment/ OJT	-	30
Total				350

Examination Centre: NIELIT Chennai, Mode: Online

Grading Scheme

Following Grading scheme (on the basis of total marks) will be followed:

Grade	S	A	B	C	D
Marks Range (in %)	$\geq 85\%$	$\geq 75\%$ and $< 85\%$	$\geq 65\%$ and $< 75\%$	$\geq 55\%$ and $< 65\%$	$\geq 50\%$ and $< 55\%$

Detailed curriculum

Module 1: Implementation of Basic AI Solution using Python programming language and SMART Framework

- SMART component and tell what each acronym means
- AI project cycle
- Orange Data Mining Tool
- An introduction to Python programming language
- Tableau Public

Module 2: Solving use cases using AI models along with building up Entrepreneurial Mindset

- Introduction to Python libraries
- AI models to solve various industry applications using Python.
- Design Thinking and AI bias
- Entrepreneurial Mindset

Module 3: Realization of Projects in AI domains with understanding of AI Project Pitfalls

- Supervised, unsupervised, and reinforcement learning
- Computer Vision, Statistical Data, Natural Language Processing and current applications of the technology

- 5 pillars of Social Emotional Skills
- AI ethics
- Project Pitfalls in relation to the AI project cycle
- IoT
- Intel's oneAPI library.

Module 4: Solving of Real time industrial problem statements using AI

- Qualify data from multiple sources
- Evaluate data for attributes
- Bias and variance
- Define and qualify AI models

